inspect_model.py — technical

`scripts/inspect_model.py · 114 lines · scripts/inspect_model.py`

Purpose

Architecture report plus an empirical check of gate application.

Step by step

Classes and functions in source order. Signatures, line ranges, raised exceptions and settings are extracted from the code; numbered steps describe what the code does.

`main`

```
def main() -> None
```

lines 33-110 · raises SystemExit

Script entry point: runs this stage end to end.

1. `bootstrap`; `load_original_model` (placement, dtype, plugin).
2. `adapter.describe(model)` for the static report; `discover_layers`.
3. Build one probe example with the plugin's task from a fixed support question/answer, collate, and run a forward pass with all layers instrumented; check the logits are finite.
4. Per layer: tokens, capacity drops, tokens per expert, mean gate, and `gate_consistency_error`.
5. `gate_ok` = max relative error below the dtype tolerance; add model source, plugin, task, placement map, environment and memory; write `architecture_report.json`.
6. Log a table of layers; exit with status 1 if the gate check failed.

Module constants

Name	Value
`PROBE_SOURCE`	'How do I reset my password if I no longer have access to my email account?'
`PROBE_TARGET`	'Contact customer support and verify your identity to recover the account.'

Errors raised

Function	Exception	Message (f-string fields in braces)
`main`	SystemExit	